

Md. Atiq Shahriar

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 [LinkedIn](#)

 [GitHub](#)

 [Portfolio](#)

Chittagong, Bangladesh

PROFESSIONAL PROFILE

Computer Science and Engineering graduate and current M.Sc. student at CUET with experience in AI/ML research and software development. Worked on federated learning and explainable AI for healthcare applications, with emerging experience in LLM application development and structured AI workflows. Skilled in Python and modern software development, with interests in AI engineering, machine learning, and applied research.

EDUCATION

Chittagong University of Engineering & Technology (CUET)

Master of Science in Computer Science & Engineering

2026 – Present

Chattogram, Bangladesh

Chittagong University of Engineering & Technology (CUET)

Bachelor of Science in Computer Science & Engineering CGPA: 3.78/4.00

March 2022 – June 2026

Chattogram, Bangladesh

Chittagong College

Higher Secondary Certificate (Science) GPA: 5.00/5.00

2018 – 2020

Chattogram, Bangladesh

Chittagong Government High School

Secondary School Certificate (Science) GPA: 5.00/5.00

2016 – 2018

Chattogram, Bangladesh

RESEARCH EXPERIENCE

Scalable and Explainable Federated Learning for Heterogeneous Electronic Health Records with Energy Profiling

2025 – 2026

Undergraduate Thesis Research, CUET

- Developed an energy-aware federated learning framework for ICU mortality prediction using MIMIC-IV data across 11 heterogeneous simulated ICU clients.
- Designed a client selection strategy that reduced training energy by **31.7%** while achieving **0.8190 AUROC** compared with full-participation FedAvg.
- Investigated the relationship between client heterogeneity, predictive performance, and training energy in federated healthcare learning.

SELECTED TECHNICAL PROJECTS

• AI Portfolio Chatbot: Chat with My AI Twin

 [GitHub](#) [Live Demo](#)

Python, FastAPI, React, Tailwind CSS, Groq API, Pydantic

- Built and deployed a full-stack chatbot where recruiters chat with an AI version of me, with streaming responses, conversation memory and resume-grounded answers that avoid hallucination.
- Added job-description matching and PDF resume upload, with LLM-based resume parsing validated by Pydantic.

• ICU Mortality Prediction Dashboard

 [GitHub](#) [Live Demo](#)

Python, PyTorch, Streamlit, LIME, Plotly, Pandas

- Built and deployed an interactive Streamlit application for ICU mortality risk prediction using a trained BiLSTM model.
- Integrated LIME-based local explanations and interactive analytics to visualize predictions, federated training performance, and energy consumption.

• MediTrack

 [GitHub](#)

React.js, Node.js, Express.js, MongoDB, Tailwind CSS, JWT, bcrypt

- Developed a full-stack healthcare platform with dedicated dashboards for doctors, patients, and administrators.
- Implemented secure authentication, role-based authorization, digital consultations, e-prescriptions, and patient history management.

• StudyPlan

 [GitHub](#)

React.js, Node.js, Express.js, MongoDB, Firebase, Tailwind CSS

- Developed a MERN-based study management platform for organizing semester-wise tasks, schedules, and academic progress.
- Built dashboards for task tracking, deadline visualization, progress monitoring, and subject-wise organization.

TECHNICAL SKILLS

- **Programming:** Python, C, C++, SQL
- **Computer Science:** Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Software Engineering
- **Artificial Intelligence & Machine Learning:** Machine Learning, Deep Learning, Federated Learning, Explainable AI
- **LLM & AI Engineering:** LLM Applications, Prompt Engineering, LLM APIs, Structured Outputs, Pydantic
- **Web Development:** React.js, Node.js, Express.js, MongoDB, REST APIs, JWT Authentication
- **Tools:** Git, GitHub, Jupyter Notebook, Google Colab, MongoDB Atlas, VS Code

LEADERSHIP & ORGANIZATIONAL EXPERIENCE

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| • Publication Secretary
<i>IEEE Computer Society CUET Student Branch Chapter</i> | 2025 – 2026 |
| • Deputy Finance Secretary
<i>CUET Computer Club</i> | 2025 – 2026 |
| • Joint Publication Secretary
<i>JOYODDHONEY, CUET</i> | 2025 – 2026 |
| • Former Head of Logistics
<i>HULT Prize at CUET</i> | 2024 – 2025 |

CERTIFICATIONS

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| Front-End Development Training (80 Hours) — CSE, CUET (EDGE Project) | Apr 2025 |
| Supervised Machine Learning: Regression and Classification — Stanford University | Mar 2025 |
| Advanced Learning Algorithms — Stanford University | Apr 2025 |
| Unsupervised Learning, Recommenders, and Reinforcement Learning — Stanford University | Jul 2025 |

REFERENCES

Prof. Dr. Mohammad Shamsul Arefin

*Professor, Department of Computer Science & Engineering
Chittagong University of Engineering & Technology (CUET)*

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